

2023 AMC 10A Problem 12 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 12. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 12

Problem:

How many three-digit positive integers N satisfy the following properties?

- ▶ The number N is divisible by 7.
- ▶ The number formed by reversing the digits of N is divisible by 5.

Answer Choices:

- A. 13
- B. 14
- C. 15
- D. 16
- E. 17

Solution:

Let M be the number formed by reversing the digits of N . Because M is divisible by 5, the last digit of M must be 0 or 5. However, the last digit of M cannot be 0, so it must be 5. It follows that N is a number between 500 and 599.

The least multiple of 7 in this range is $504 = 7 \cdot 72$, while the greatest multiple of 7 in this range is $595 = 7 \cdot 85$. Thus N is one of the numbers $7 \cdot 72, 7 \cdot 73, 7 \cdot 74, \dots$, or $7 \cdot 85$. There are $85 - 72 + 1 = \text{(B)} \boxed{14}$ numbers in this list.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)